loan23-03-data-cleaning-eda-catboost

March 21, 2026

1 Loan Approval Prediction - CPE 232 Data Models

This notebook is a step-by-step Kaggle workflow for the **Loan Approval Prediction** competition.

Notebook goals 1. Load the competition data in a Kaggle-native way. 2. Validate the schema so we know training and submission logic are aligned. 3. Clean the raw columns and document what changed. 4. Run a real EDA section before modeling. 5. Train stronger CatBoost CV baselines with threshold tuning. 6. Export artifacts and a submission file using the new notebook slug.

1.1 Workflow

We will move in the same order a human notebook usually does:

- setup packages and configuration
- resolve data paths and validate inputs
- inspect data quality and summarize cleaning actions
- run EDA and write down takeaways
- engineer stronger features
- compare feature bundles with CatBoost CV
- choose a threshold based on OOF F1
- export artifacts and prepare the submission file

[1]:
```
# Cell: Setup - install missing packages, then import the notebook stack
import importlib.util
import subprocess
import sys


def run_cmd(cmd):
    print("Running:", " ".join(cmd))
    subprocess.check_call(cmd)


def ensure_package(module_name, package_name=None):
    pkg = package_name or module_name
    if importlib.util.find_spec(module_name) is None:
        run_cmd([sys.executable, "-m", "pip", "install", "-q", pkg])
```

```
ensure_package("numpy")
ensure_package("pandas")
ensure_package("matplotlib")
ensure_package("seaborn")
ensure_package("sklearn", "scikit-learn")
ensure_package("catboost")
ensure_package("kaggle")
ensure_package("dotenv", "python-dotenv")
```

[2]:
```
# Cell: Imports - bring in all libraries used across EDA, training, and␣
↪submission
import os
import time
import warnings
from datetime import UTC, datetime
from pathlib import Path
from zipfile import ZipFile

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import seaborn as sns
from catboost import CatBoostClassifier
from dotenv import load_dotenv
from IPython.display import display
from kaggle.api.kaggle_api_extended import KaggleApi
from sklearn.metrics import (
    confusion_matrix,
    f1_score,
    precision_score,
    recall_score,
)
from sklearn.model_selection import StratifiedKFold

warnings.filterwarnings("ignore")
pd.set_option("display.max_columns", 200)
pd.set_option("display.float_format", lambda x: f"{x:,.4f}")
sns.set_style("whitegrid")
plt.rcParams["figure.figsize"] = (14, 6)

print("Imports ready.")
```

```
Imports ready.
```

1.2 Notebook Config

This section keeps the same runtime surface as the fast notebook, so it still works well on Kaggle and still produces the same kinds of outputs. The difference is that we will inspect more before training.

[ ]:
```
# Cell: Config - notebook identity, runtime flags, and competition schema
COMPETITION = "insurance-premium-prediction-cpe-232-data-models"
NOTEBOOK_SLUG = "loan23-03-human-eda-gpu-fast"

RUN_DOWNLOAD = True
RUN_SUBMISSION = True
FORCE_DOWNLOAD = True

CV_MODE = "stratified"
N_SPLITS = 5
SEED = 42
ENABLE_OPTUNA = False
ENABLE_ABLATIONS = True
ENABLE_ENSEMBLE = False

CATBOOST_ITERATIONS = 1500
EARLY_STOPPING_ROUNDS = 100
USE_GPU = True
GPU_DEVICE = "0"

ID_COL = "id"
TARGET_COL = "loan_status"
CATEGORICAL_COLS = ["education", "self_employed"]
NUMERIC_COLS = [
    "no_of_dependents",
    "income_annum",
    "loan_amount",
    "loan_term",
    "cibil_score",
    "residential_assets_value",
    "commercial_assets_value",
    "luxury_assets_value",
    "bank_asset_value",
]
MONEY_COLS = [
    "income_annum",
    "loan_amount",
    "residential_assets_value",
    "commercial_assets_value",
    "luxury_assets_value",
    "bank_asset_value",
]
```

```python
def detect_gpu_available() -> bool:
    if not USE_GPU:
        return False
    try:
        proc = subprocess.run(
            ["nvidia-smi", "-L"],
            capture_output=True,
            text=True,
            timeout=5,
            check=False,
        )
        return proc.returncode == 0 and "GPU" in proc.stdout
    except Exception:
        return False


GPU_AVAILABLE = detect_gpu_available()
TRAIN_DEVICE_LABEL = "gpu" if GPU_AVAILABLE else "cpu"
OUTPUT_ROOT = (
    Path("/kaggle/working") if Path("/kaggle/working").exists() else Path.cwd()
)
ARTIFACT_DIR = OUTPUT_ROOT / "outputs"
ARTIFACT_DIR.mkdir(parents=True, exist_ok=True)

print("Competition      :", COMPETITION)
print("Notebook slug    :", NOTEBOOK_SLUG)
print("Training device  :", TRAIN_DEVICE_LABEL)
print("CV mode / folds  :", CV_MODE, N_SPLITS)
print("Artifacts dir    :", ARTIFACT_DIR)
```

```
Competition      : insurance-premium-prediction-cpe-232-data-models
Notebook slug    : loan23-03-human-eda-gpu-fast
Training device  : gpu
CV mode / folds  : stratified 5
Artifacts dir    : /kaggle/working/outputs
```

1.3 Data Resolution

We want one notebook that works in both places:

- locally, if the CSV files already exist in the project folder
- on Kaggle, if the competition files are mounted or need to be downloaded

That means we resolve the best available data path first, then validate the files before touching modeling code.

[4]:
```python
# Cell: Data access - resolve local or Kaggle data and authenticate if possible
def load_env_from_candidates():
    cwd = Path.cwd().resolve()
    candidates = [cwd / ".env", cwd.parent / ".env", Path("/kaggle/working/.
 ↪env")]
    loaded = False
    for env_path in candidates:
        if env_path.exists():
            load_dotenv(env_path, override=False)
            print(f"Loaded env from {env_path}")
            loaded = True
            break
    if not loaded:
        print("No .env file found in default candidate locations.")


def resolve_local_data_dir() -> Path:
    cwd = Path.cwd().resolve()
    candidates = [
        cwd / "data",
        cwd / "notebooks/
 ↪2026-03-20-loan-approval-prediction-cpe-232-data-models/data",
        cwd.parent / "data",
    ]
    for c in candidates:
        if c.exists() and (c / "train.csv").exists() and (c / "test.csv").
 ↪exists():
            return c
    return cwd / "data"


def resolve_kaggle_input_dir(competition: str) -> Path | None:
    direct = Path("/kaggle/input") / competition
    if direct.exists() and (direct / "train.csv").exists():
        return direct

    if Path("/kaggle/input").exists():
        matches = sorted(Path("/kaggle/input").glob(f"*{competition}*"))
        for m in matches:
            if (m / "train.csv").exists() and (m / "test.csv").exists():
                return m
    return None


def ensure_competition_data(
    api: KaggleApi | None, competition: str, run_download: bool, force_download:
 ↪ bool
```

```
):
    kaggle_input = resolve_kaggle_input_dir(competition)
    if kaggle_input is not None and not run_download:
        print("Using mounted Kaggle input:", kaggle_input)
        return kaggle_input

    local_data = resolve_local_data_dir()
    local_data.mkdir(parents=True, exist_ok=True)

    train_path = local_data / "train.csv"
    test_path = local_data / "test.csv"
    sample_path = local_data / "sample_submission.csv"
    have_local = train_path.exists() and test_path.exists() and sample_path.
↪exists()

    if run_download:
        if api is None:
            if not have_local:
                raise RuntimeError(
                    "RUN_DOWNLOAD=True but Kaggle auth unavailable and no local␣
↪data found."
                )
            print("RUN_DOWNLOAD=True but API unavailable. Falling back to local␣
↪files.")
            return local_data

        if force_download and local_data.exists():
            for z in local_data.glob("*.zip"):
                try:
                    z.unlink()
                except Exception:
                    pass

        if force_download or not have_local:
            print("Downloading competition files via Kaggle API...")
            api.competition_download_files(
                competition, path=str(local_data), force=force_download,␣
↪quiet=False
            )
            zip_candidates = sorted(local_data.glob("*.zip"))
            if not zip_candidates:
                raise RuntimeError("Kaggle download did not produce a zip file.
↪")
            latest_zip = max(zip_candidates, key=lambda p: p.stat().st_mtime)
            with ZipFile(latest_zip, "r") as zf:
                zf.extractall(local_data)
            print("Extracted:", latest_zip.name)
```

```
        have_local = train_path.exists() and test_path.exists() and sample_path.
↪exists()
        if not have_local:
            raise RuntimeError(f"Data missing after download in {local_data}")

    if not have_local:
        raise RuntimeError(
            f"Data not found. Checked local dir {local_data}. "
            "Set RUN_DOWNLOAD=True with valid Kaggle credentials or provide␣
↪local CSV files."
        )

    return local_data


load_env_from_candidates()
os.environ.pop("KAGGLE_API_TOKEN", None)

api = None
try:
    api = KaggleApi()
    api.authenticate()
    print("Authenticated user:", api.get_config_value("username"))
except Exception as auth_exc:
    print("Kaggle auth unavailable:", auth_exc)

data_dir = ensure_competition_data(
    api=api,
    competition=COMPETITION,
    run_download=RUN_DOWNLOAD,
    force_download=FORCE_DOWNLOAD,
)
print("Resolved data dir:", data_dir)
```

```
No .env file found in default candidate locations.
Authenticated user: beambeambeam
Downloading competition files via Kaggle API…
Downloading insurance-premium-prediction-cpe-232-data-models.zip to
/kaggle/working/data

100%|      | 1.24M/1.24M [00:00<00:00, 120MB/s]


Extracted: insurance-premium-prediction-cpe-232-data-models.zip
Resolved data dir: /kaggle/working/data
```

1.4 Schema Validation

Before we clean or model anything, we verify that:

- train/test/sample files exist
- columns appear in the exact expected order
- numeric columns can be parsed
- the target is binary

This keeps downstream debugging much easier.

[ ]:
```
# Cell: Load data - validate schema, parse numeric columns, and inspect the␣
↪target
EXPECTED_TRAIN_COLUMNS = [
    "id",
    "no_of_dependents",
    "education",
    "self_employed",
    "income_annum",
    "loan_amount",
    "loan_term",
    "cibil_score",
    "residential_assets_value",
    "commercial_assets_value",
    "luxury_assets_value",
    "bank_asset_value",
    "loan_status",
]
EXPECTED_TEST_COLUMNS = [
    "id",
    "no_of_dependents",
    "education",
    "self_employed",
    "income_annum",
    "loan_amount",
    "loan_term",
    "cibil_score",
    "residential_assets_value",
    "commercial_assets_value",
    "luxury_assets_value",
    "bank_asset_value",
]


def fail_schema(message: str):
    raise ValueError(f"[SCHEMA ERROR] {message}")


def load_and_validate_data(data_dir: Path):
```

```
    train_path = data_dir / "train.csv"
    test_path = data_dir / "test.csv"
    sample_path = data_dir / "sample_submission.csv"

    if not train_path.exists() or not test_path.exists() or not sample_path.
↪exists():
        fail_schema("Expected train.csv, test.csv, and sample_submission.csv to␣
↪exist.")

    train_df = pd.read_csv(train_path)
    test_df = pd.read_csv(test_path)
    sample_df = pd.read_csv(sample_path)

    train_df.columns = [str(c).strip() for c in train_df.columns]
    test_df.columns = [str(c).strip() for c in test_df.columns]
    sample_df.columns = [str(c).strip() for c in sample_df.columns]

    if train_df.columns.tolist() != EXPECTED_TRAIN_COLUMNS:
        fail_schema(
            f"Unexpected train columns. expected={EXPECTED_TRAIN_COLUMNS},␣
↪got={train_df.columns.tolist()}"
        )
    if test_df.columns.tolist() != EXPECTED_TEST_COLUMNS:
        fail_schema(
            f"Unexpected test columns. expected={EXPECTED_TEST_COLUMNS},␣
↪got={test_df.columns.tolist()}"
        )
    if sample_df.columns.tolist() != [ID_COL, TARGET_COL]:
        fail_schema(
            f"Unexpected sample_submission columns. expected={[ID_COL,␣
↪TARGET_COL]}, got={sample_df.columns.tolist()}"
        )

    train_features = [c for c in train_df.columns if c != TARGET_COL]
    if train_features != test_df.columns.tolist():
        fail_schema("Train/test feature mismatch after removing target column.")

    for col in NUMERIC_COLS + [TARGET_COL]:
        train_df[col] = pd.to_numeric(train_df[col], errors="coerce")
    for col in NUMERIC_COLS:
        test_df[col] = pd.to_numeric(test_df[col], errors="coerce")

    if not set(train_df[TARGET_COL].dropna().unique()).issubset({0, 1}):
        fail_schema("Target column must contain only 0/1 values.")

    print("Train shape :", train_df.shape)
    print("Test shape  :", test_df.shape)
```

```
    print("Sample shape:", sample_df.shape)
    print()
    print("Target balance:")
    display(
        train_df[TARGET_COL].value_counts(normalize=True).sort_index().
 ↪rename("ratio")
    )

    return train_df, test_df, sample_df


train_df, test_df, sample_submission = load_and_validate_data(data_dir)
print("Schema ready.")
```

```
Train shape : (32000, 13)
Test shape  : (8000, 12)
Sample shape: (8000, 2)

Target balance:

loan_status
0   0.4553
1   0.5447
Name: ratio, dtype: float64

Schema ready.
```

1.5 Data Cleaning Log

Now we inspect what needs normalization. Even if the dataset is already clean, it is still helpful to make the cleaning rules explicit so the feature pipeline is reproducible.

[6]:
```
# Cell: Cleaning prep - normalize raw fields and record what changed
MISSING_TOKENS = {"", "na", "n/a", "none", "null", "nan", "unknown", "missing",␣
 ↪"?"}


def normalize_text_like(series: pd.Series) -> pd.Series:
    s = series.astype("string").str.replace(r"\s+", " ", regex=True).str.strip()
    low = s.str.lower()
    return s.mask(low.isin(MISSING_TOKENS), pd.NA)


def robust_numeric(series: pd.Series) -> pd.Series:
    out = pd.to_numeric(series, errors="coerce")
    return out.replace([np.inf, -np.inf], np.nan)


def clean_base_frames(train_raw: pd.DataFrame, test_raw: pd.DataFrame):
```

```
    tr = train_raw.copy(deep=True)
    te = test_raw.copy(deep=True)

    for col in NUMERIC_COLS:
        tr[col] = robust_numeric(tr[col])
        te[col] = robust_numeric(te[col])

    tr[TARGET_COL] = robust_numeric(tr[TARGET_COL]).astype("int8")

    for col in CATEGORICAL_COLS:
        tr[col] = normalize_text_like(tr[col])
        te[col] = normalize_text_like(te[col])

    return tr, te


train_clean_base, test_clean_base = clean_base_frames(train_df, test_df)

cleaning_log = pd.DataFrame(
    [
        {
            "check": "missing_train_rows",
            "value": int(train_df.isna().any(axis=1).sum()),
            "notes": "Rows with at least one missing value in the training set.
 ↪",
        },
        {
            "check": "missing_test_rows",
            "value": int(test_df.isna().any(axis=1).sum()),
            "notes": "Rows with at least one missing value in the test set.",
        },
        {
            "check": "train_duplicates",
            "value": int(train_df.duplicated().sum()),
            "notes": "Exact duplicate rows in train.",
        },
        {
            "check": "test_duplicates",
            "value": int(test_df.duplicated().sum()),
            "notes": "Exact duplicate rows in test.",
        },
        {
            "check": "normalized_categories",
            "value": len(CATEGORICAL_COLS),
            "notes": "Text columns trimmed, case-normalized, and missing-like␣
 ↪tokens standardized.",
        },
```

```
    ]
)

display(cleaning_log)

missing_summary = pd.concat(
    [
        train_df.isna().sum().rename("missing_train"),
        train_df.isna().mean().mul(100).rename("missing_train_pct"),
        test_df.isna().sum().rename("missing_test"),
        test_df.isna().mean().mul(100).rename("missing_test_pct"),
    ],
    axis=1,
).sort_values(["missing_train", "missing_test"], ascending=False)

display(missing_summary)
```

```
                   check  value  \
0     missing_train_rows  17289
1      missing_test_rows   4348
2       train_duplicates      0
3        test_duplicates      0
4  normalized_categories      2

                                                notes
0  Rows with at least one missing value in the tr…
1  Rows with at least one missing value in the te…
2                     Exact duplicate rows in train.
3                      Exact duplicate rows in test.
4  Text columns trimmed, case-normalized, and mis…

                         missing_train  missing_train_pct  missing_test  \
loan_amount                       2918             9.1188      732.0000
cibil_score                       2770             8.6562      706.0000
self_employed                     2623             8.1969      652.0000
residential_assets_value          2619             8.1844      694.0000
income_annum                      2135             6.6719      518.0000
commercial_assets_value           2097             6.5531      489.0000
bank_asset_value                  1918             5.9937      475.0000
loan_term                         1897             5.9281      490.0000
luxury_assets_value               1834             5.7313      487.0000
education                         1737             5.4281      455.0000
no_of_dependents                  1605             5.0156      421.0000
id                                   0             0.0000        0.0000
loan_status                          0             0.0000           NaN

                         missing_test_pct
loan_amount                        9.1500
```

```
cibil_score                        8.8250
self_employed                      8.1500
residential_assets_value           8.6750
income_annum                       6.4750
commercial_assets_value            6.1125
bank_asset_value                   5.9375
loan_term                          6.1250
luxury_assets_value                6.0875
education                          5.6875
no_of_dependents                   5.2625
id                                 0.0000
loan_status                           NaN
```

1.6 EDA — Overview

Before feature engineering, we want quick answers to a few practical questions:

- Is the target balanced or skewed?
- Are there obvious missing-value or duplicate issues?
- Which variables look most related to approval?
- Are any numeric features heavily skewed?

We will answer those with small, readable cells instead of one giant block.

[ ]:
```
# Cell: EDA overview - shape, preview, target balance, duplicates, and summary␣
↪statistics
print("Train preview:")
display(train_df.head())

print("Test preview:")
display(test_df.head())

overview_df = pd.DataFrame(
    {
        "dataset": ["train", "test"],
        "rows": [len(train_df), len(test_df)],
        "columns": [train_df.shape[1], test_df.shape[1]],
        "duplicate_rows": [
            int(train_df.duplicated().sum()),
            int(test_df.duplicated().sum()),
        ],
    }
)
display(overview_df)

target_summary = (
    train_df[TARGET_COL]
    .value_counts(dropna=False)
    .rename_axis(TARGET_COL)
```

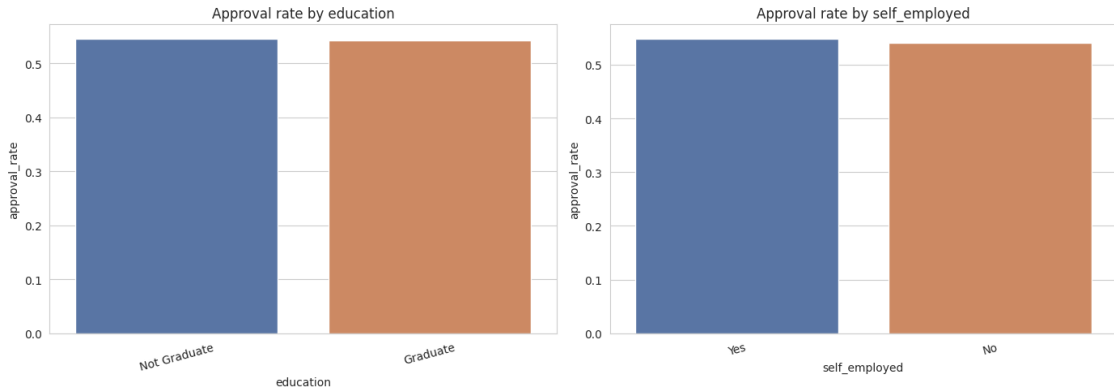
```
    .rename("count")
    .reset_index()
)
target_summary["ratio"] = target_summary["count"] / target_summary["count"].
 ↪sum()
display(target_summary)

numeric_summary = train_df[NUMERIC_COLS].describe().T
display(numeric_summary)
```

```
Train preview:

   id  no_of_dependents      education self_employed   income_annum  \
0   1            0.0000       Graduate            No            NaN
1   2            2.0000   Not Graduate           Yes 2,832,292.0000
2   3            2.0000       Graduate            No 2,594,879.0000
3   4            4.0000       Graduate            No 2,497,120.0000
4   5            5.0000   Not Graduate           Yes            NaN

      loan_amount  loan_term  cibil_score  residential_assets_value  \
0  7,271,300.0000     5.0000     622.0000            6,572,296.0000
1             NaN     9.0000     482.0000            3,274,507.0000
2  8,265,934.0000    16.0000     786.0000              179,550.0000
3 26,116,812.0000    16.0000     352.0000           16,697,499.0000
4  4,071,241.0000    19.0000     381.0000                      NaN

   commercial_assets_value  luxury_assets_value  bank_asset_value  loan_status
0          12,264,202.0000                  NaN    4,471,627.0000            0
1          16,079,677.0000       2,643,225.0000    3,021,958.0000            1
2           1,361,037.0000       9,057,816.0000               NaN            0
3                      NaN               0.0000      983,214.0000            0
4           2,788,429.0000       7,977,004.0000   11,154,770.0000            1

Test preview:

      id  no_of_dependents      education self_employed   income_annum  \
0  32001            1.0000       Graduate           Yes 1,184,682.0000
1  32002            0.0000       Graduate            No 8,920,224.0000
2  32003            0.0000   Not Graduate           Yes 9,064,660.0000
3  32004            5.0000       Graduate            No 9,787,483.0000
4  32005            1.0000   Not Graduate            No   320,746.0000

      loan_amount  loan_term  cibil_score  residential_assets_value  \
0 17,119,708.0000    11.0000     749.0000                    0.0000
1  5,040,529.0000     0.0000     989.0000                       NaN
2 15,390,179.0000    14.0000     511.0000            1,254,869.0000
3  5,649,434.0000        NaN     513.0000            1,315,339.0000
4  2,157,362.0000    13.0000     759.0000            3,273,669.0000
```

```
   commercial_assets_value  luxury_assets_value  bank_asset_value
0           4,800,216.0000      16,565,306.0000    1,631,095.0000
1           3,590,433.0000      22,490,251.0000    4,153,781.0000
2           6,981,191.0000      12,755,544.0000    1,497,759.0000
3             421,501.0000      21,345,655.0000      945,687.0000
4           2,587,980.0000      25,049,850.0000    1,374,027.0000

  dataset   rows  columns  duplicate_rows
0   train  32000       13               0
1    test   8000       12               0

   loan_status  count  ratio
0            1  17430 0.5447
1            0  14570 0.4553

                              count            mean            std      min  \
no_of_dependents         30,395.0000          1.9519         1.6312   0.0000
income_annum             29,865.0000  3,523,465.8867 2,920,311.3126   0.0000
loan_amount              29,082.0000 12,851,018.5821 9,177,296.6448   0.0000
loan_term                30,103.0000         12.3672         5.4403   0.0000
cibil_score              29,230.0000        661.7312       173.0936 223.0000
residential_assets_value 29,381.0000  3,956,637.7345 5,339,316.2166   0.0000
commercial_assets_value  29,903.0000  6,212,830.1276 5,490,717.1960   0.0000
luxury_assets_value      30,166.0000 11,195,238.6189 8,793,568.2731   0.0000
bank_asset_value         30,082.0000  4,702,795.4577 3,770,260.7589   0.0000

                                  25%             50%             75%  \
no_of_dependents               0.0000          2.0000          3.0000
income_annum             830,284.0000  2,961,480.0000  5,827,615.0000
loan_amount            5,416,666.0000 11,746,169.0000 18,701,166.0000
loan_term                      9.0000         13.0000         17.0000
cibil_score                  529.0000        663.5000        811.0000
residential_assets_value       0.0000  1,812,942.0000  5,528,413.0000
commercial_assets_value  1,880,555.0000  4,531,977.0000  9,059,979.5000
luxury_assets_value      4,067,125.2500  9,802,424.5000 16,711,702.7500
bank_asset_value         1,596,127.2500  3,998,887.5000  7,027,443.7500

                                   max
no_of_dependents                5.0000
income_annum            11,369,423.0000
loan_amount             46,044,308.0000
loan_term                       22.0000
cibil_score                  1,011.0000
residential_assets_value 34,562,020.0000
commercial_assets_value  24,041,102.0000
luxury_assets_value      46,505,087.0000
bank_asset_value         17,479,340.0000
```

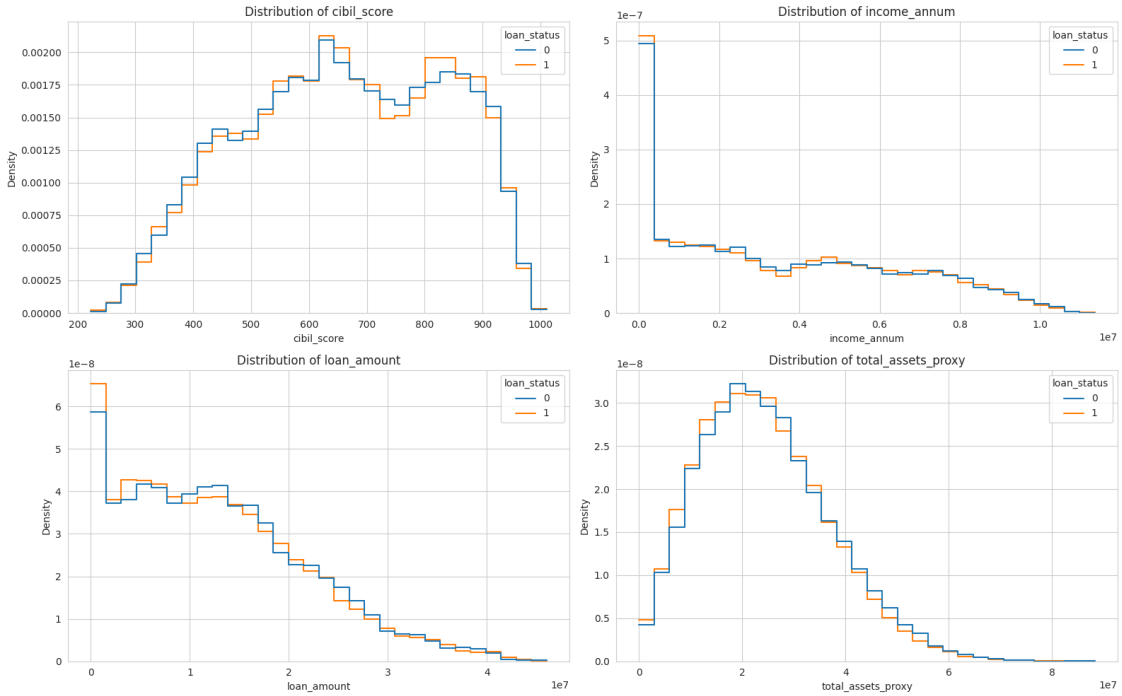
[8]:
```python
# Cell: EDA categorical review - category mix and approval rate by category
for col in CATEGORICAL_COLS:
    print(f"Column: {col}")
    count_df = (
        train_clean_base.groupby(col, dropna=False)[TARGET_COL]
        .agg(["count", "mean"])
        .rename(columns={"mean": "approval_rate"})
        .sort_values(["approval_rate", "count"], ascending=[False, False])
        .reset_index()
    )
    display(count_df)

fig, axes = plt.subplots(1, len(CATEGORICAL_COLS), figsize=(14, 5))
if len(CATEGORICAL_COLS) == 1:
    axes = [axes]

for ax, col in zip(axes, CATEGORICAL_COLS):
    rate_df = (
        train_clean_base.groupby(col, dropna=False)[TARGET_COL]
        .mean()
        .sort_values(ascending=False)
        .rename("approval_rate")
        .reset_index()
    )
    sns.barplot(data=rate_df, x=col, y="approval_rate", ax=ax, palette="deep")
    ax.set_title(f"Approval rate by {col}")
    ax.set_xlabel(col)
    ax.set_ylabel("approval_rate")
    ax.tick_params(axis="x", rotation=15)

plt.tight_layout()
plt.show()
```

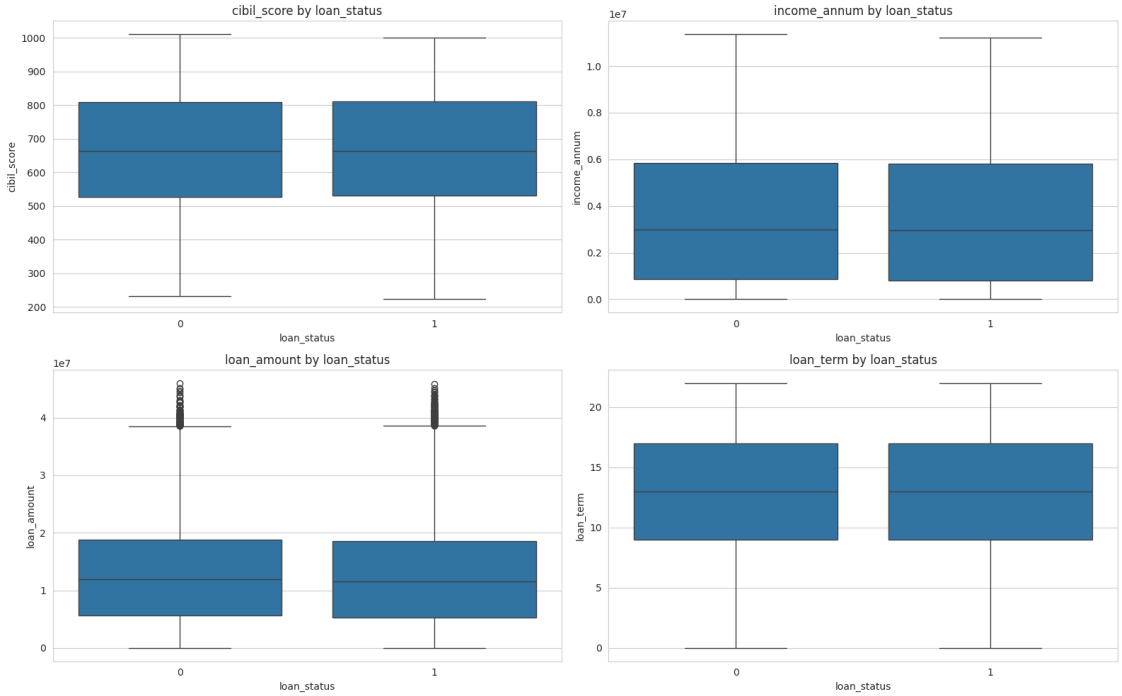
```
Column: education

      education  count  approval_rate
0          <NA>   1737         0.5527
1  Not Graduate  16081         0.5451
2      Graduate  14182         0.5433

Column: self_employed

  self_employed  count  approval_rate
0          <NA>   2623         0.5505
1           Yes  15939         0.5473
2            No  13438         0.5404
```



[9]:
```
# Cell: EDA numeric distributions - inspect scale, skew, and approval separation
plot_cols = ["cibil_score", "income_annum", "loan_amount", "total_assets_proxy"]
plot_df = train_clean_base.copy()
plot_df["total_assets_proxy"] = (
    plot_df["residential_assets_value"].fillna(0)
    + plot_df["commercial_assets_value"].fillna(0)
    + plot_df["luxury_assets_value"].fillna(0)
    + plot_df["bank_asset_value"].fillna(0)
)

fig, axes = plt.subplots(2, 2, figsize=(16, 10))
for ax, col in zip(axes.flatten(), plot_cols):
    sns.histplot(
        data=plot_df,
        x=col,
        hue=TARGET_COL,
        bins=30,
        stat="density",
        common_norm=False,
        element="step",
        fill=False,
        ax=ax,
    )
    ax.set_title(f"Distribution of {col}")
plt.tight_layout()
plt.show()

box_cols = ["cibil_score", "income_annum", "loan_amount", "loan_term"]
fig, axes = plt.subplots(2, 2, figsize=(16, 10))
for ax, col in zip(axes.flatten(), box_cols):
    sns.boxplot(data=train_clean_base, x=TARGET_COL, y=col, ax=ax)
    ax.set_title(f"{col} by loan_status")
plt.tight_layout()
```

```
plt.show()
```

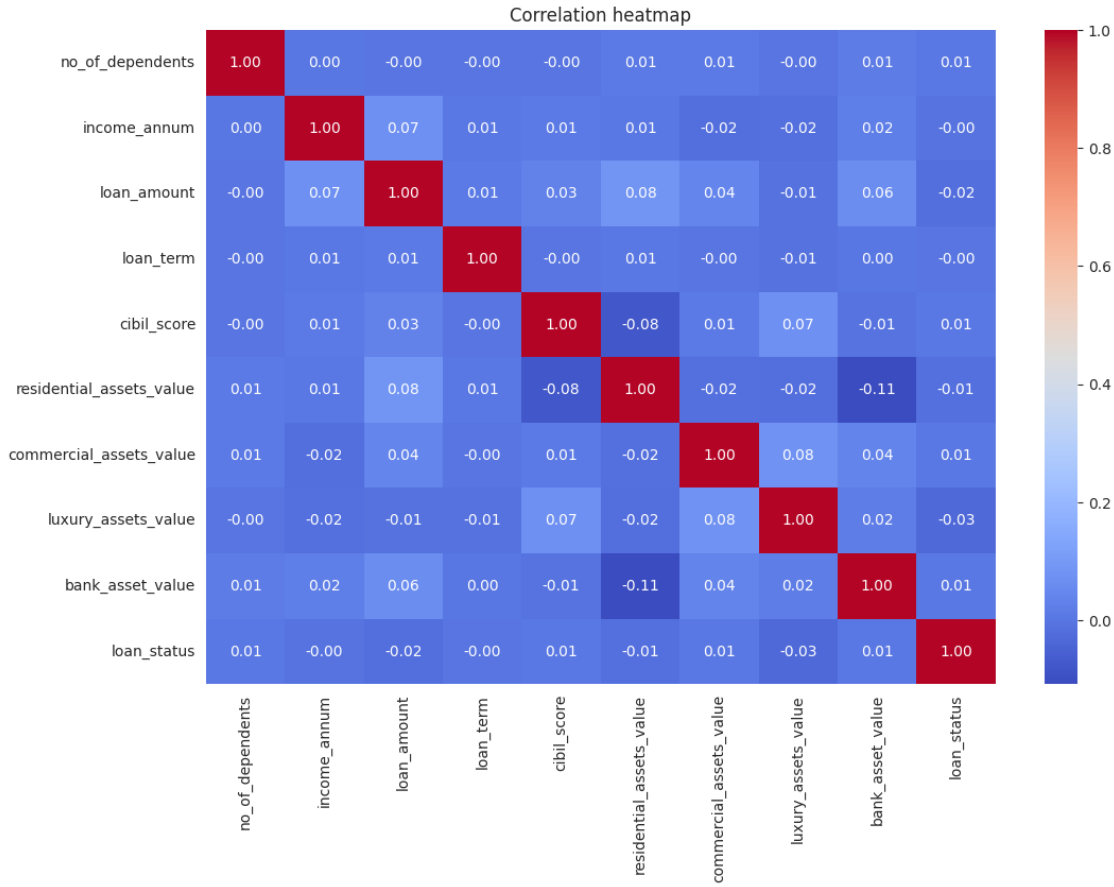

Distribution of cibil_score
Density
cibil_score
loan_status
0
1
Distribution of income_annum
1e−7
Density
income_annum
1e7
Distribution of loan_amount
1e−8
Density
loan_amount
1e7
Distribution of total_assets_proxy
1e−8
Density
total_assets_proxy
1e7


cibil_score by loan_status
cibil_score
loan_status
income_annum by loan_status
1e7
income_annum
loan_status
loan_amount by loan_status
1e7
loan_amount
loan_status
loan_term by loan_status
loan_term
loan_status

[ ]:
```
# Cell: EDA relationships - approval rates across buckets and numeric␣
↪correlation
bucket_df = train_clean_base.copy()
bucket_df["cibil_band_preview"] = pd.cut(
    bucket_df["cibil_score"],
    bins=[-np.inf, 500, 650, 750, 850, np.inf],
    labels=["very_low", "low", "mid", "high", "very_high"],
)
bucket_df["loan_to_income_preview"] = bucket_df["loan_amount"] / bucket_df[
    "income_annum"
].replace(0, np.nan)
bucket_df["loan_to_income_bucket"] = pd.qcut(
    bucket_df["loan_to_income_preview"].fillna(
        bucket_df["loan_to_income_preview"].median()
    ),
    q=4,
    duplicates="drop",
)

approval_by_band = (
    bucket_df.groupby("cibil_band_preview", dropna=False)[TARGET_COL]
    .mean()
    .rename("approval_rate")
    .reset_index()
)
display(approval_by_band)

approval_by_ratio = (
    bucket_df.groupby("loan_to_income_bucket", dropna=False)[TARGET_COL]
    .mean()
    .rename("approval_rate")
    .reset_index()
)
display(approval_by_ratio)

corr_cols = NUMERIC_COLS + [TARGET_COL]
corr_df = train_df[corr_cols].corr(numeric_only=True)
plt.figure(figsize=(12, 8))
sns.heatmap(corr_df, annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Correlation heatmap")
plt.show()

eda_takeaways = pd.DataFrame(
    [
        {
            "takeaway": "credit_score_signal",
```

```
            "detail": "CIBIL score shows a strong directional relationship with␣
↪approval and deserves both raw and bucketed treatment.",
        },
        {
            "takeaway": "scale_skew",
            "detail": "Income, loan amount, and asset values are skewed, so log␣
↪features should help the model.",
        },
        {
            "takeaway": "ratio_features",
            "detail": "Loan-to-income and asset-to-loan style ratios look␣
↪meaningful for separating approved vs rejected applications.",
        },
    ]
)
display(eda_takeaways)
```

```
  cibil_band_preview  approval_rate
0           very_low         0.5396
1                low         0.5467
2                mid         0.5445
3               high         0.5496
4          very_high         0.5471
5                NaN         0.5372

  loan_to_income_bucket  approval_rate
0       (-0.001, 1.963]         0.5541
1        (1.963, 3.101]         0.5477
2        (3.101, 5.041]         0.5337
3     (5.041, 37754.78]         0.5357
```




```
              takeaway                                                 detail
0  credit_score_signal  CIBIL score shows a strong directional relatio…
1            scale_skew  Income, loan amount, and asset values are skew…
2       ratio_features  Loan-to-income and asset-to-loan style ratios …
```

1.7 Feature Engineering

Now we turn the EDA observations into reusable feature bundles. We keep the same three variants from the fast notebook:

- `raw_minimal`
- `processed_basic`
- `processed_extended`

That lets us compare how much value each layer of feature work adds.

[ ]:
```
# Cell: Feature engineering helpers - imputation, indicators, ratios, logs, and␣
↪category typing
FEATURE_CACHE = {}
```

```python
def fill_numeric_median(tr: pd.DataFrame, te: pd.DataFrame, columns: list[str]):
    for col in columns:
        med = float(tr[col].median()) if tr[col].notna().any() else 0.0
        tr[col] = tr[col].fillna(med)
        te[col] = te[col].fillna(med)


def fill_categorical_missing(tr: pd.DataFrame, te: pd.DataFrame, columns:␣
 ↪list[str]):
    for col in columns:
        tr[col] = tr[col].fillna("Missing").astype("string")
        te[col] = te[col].fillna("Missing").astype("string")

        vocab = sorted(set(tr[col].dropna().astype(str).unique()))
        if "Unknown" not in vocab:
            vocab.append("Unknown")
        tr[col] = tr[col].where(tr[col].isin(vocab), "Unknown")
        te[col] = te[col].where(te[col].isin(vocab), "Unknown")


def add_missing_indicators(tr: pd.DataFrame, te: pd.DataFrame, columns:␣
 ↪list[str]):
    created = []
    for col in columns:
        if tr[col].isna().mean() <= 0:
            continue
        name = f"mis_{col}"
        tr[name] = tr[col].isna().astype("int8")
        te[name] = te[col].isna().astype("int8")
        created.append(name)
    return created


def add_extended_features(tr: pd.DataFrame, te: pd.DataFrame):
    asset_cols = [
        "residential_assets_value",
        "commercial_assets_value",
        "luxury_assets_value",
        "bank_asset_value",
    ]

    tr["total_assets"] = tr[asset_cols].sum(axis=1)
    te["total_assets"] = te[asset_cols].sum(axis=1)

    loan_safe_tr = tr["loan_amount"].replace(0, np.nan)
    loan_safe_te = te["loan_amount"].replace(0, np.nan)
    income_safe_tr = tr["income_annum"].replace(0, np.nan)
```

```
income_safe_te = te["income_annum"].replace(0, np.nan)
term_safe_tr = tr["loan_term"].replace(0, np.nan)
term_safe_te = te["loan_term"].replace(0, np.nan)

tr["total_assets_to_loan"] = (tr["total_assets"] / loan_safe_tr).replace(
    [np.inf, -np.inf], np.nan
)
te["total_assets_to_loan"] = (te["total_assets"] / loan_safe_te).replace(
    [np.inf, -np.inf], np.nan
)
tr["income_to_loan"] = (tr["income_annum"] / loan_safe_tr).replace(
    [np.inf, -np.inf], np.nan
)
te["income_to_loan"] = (te["income_annum"] / loan_safe_te).replace(
    [np.inf, -np.inf], np.nan
)
tr["loan_to_income"] = (tr["loan_amount"] / income_safe_tr).replace(
    [np.inf, -np.inf], np.nan
)
te["loan_to_income"] = (te["loan_amount"] / income_safe_te).replace(
    [np.inf, -np.inf], np.nan
)
tr["loan_per_term"] = (tr["loan_amount"] / term_safe_tr).replace(
    [np.inf, -np.inf], np.nan
)
te["loan_per_term"] = (te["loan_amount"] / term_safe_te).replace(
    [np.inf, -np.inf], np.nan
)
tr["income_per_dependent"] = tr["income_annum"] / (
    tr["no_of_dependents"].fillna(0) + 1.0
)
te["income_per_dependent"] = te["income_annum"] / (
    te["no_of_dependents"].fillna(0) + 1.0
)
tr["assets_minus_loan"] = tr["total_assets"] - tr["loan_amount"].fillna(0)
te["assets_minus_loan"] = te["total_assets"] - te["loan_amount"].fillna(0)

total_assets_safe_tr = tr["total_assets"].replace(0, np.nan)
total_assets_safe_te = te["total_assets"].replace(0, np.nan)
for col in asset_cols:
    ratio_name = f"{col}_share"
    tr[ratio_name] = (tr[col] / total_assets_safe_tr).replace(
        [np.inf, -np.inf], np.nan
    )
    te[ratio_name] = (te[col] / total_assets_safe_te).replace(
        [np.inf, -np.inf], np.nan
    )
```

```
    for col in MONEY_COLS + ["total_assets"]:
        tr[f"log1p_{col}"] = np.log1p(np.clip(tr[col].fillna(0), a_min=0,␣
↪a_max=None))
        te[f"log1p_{col}"] = np.log1p(np.clip(te[col].fillna(0), a_min=0,␣
↪a_max=None))

    tr["cibil_band"] = pd.cut(
        tr["cibil_score"],
        bins=[-np.inf, 500, 650, 750, 850, np.inf],
        labels=["very_low", "low", "mid", "high", "very_high"],
    ).astype("string")
    te["cibil_band"] = pd.cut(
        te["cibil_score"],
        bins=[-np.inf, 500, 650, 750, 850, np.inf],
        labels=["very_low", "low", "mid", "high", "very_high"],
    ).astype("string")


def finalize_frame_types(tr: pd.DataFrame, te: pd.DataFrame, feature_cols:␣
↪list[str]):
    cat_cols = []
    for col in feature_cols:
        if str(tr[col].dtype) in {"object", "string", "category"}:
            tr[col] = tr[col].astype("string")
            te[col] = te[col].astype("string")
            cat_cols.append(col)
        else:
            tr[col] = robust_numeric(tr[col]).astype("float32")
            te[col] = robust_numeric(te[col]).astype("float32")
    return cat_cols


def build_features(variant: str, train_raw: pd.DataFrame, test_raw: pd.
↪DataFrame):
    if variant not in {"raw_minimal", "processed_basic", "processed_extended"}:
        raise ValueError(f"Unknown feature variant: {variant}")

    tr, te = clean_base_frames(train_raw, test_raw)

    if variant == "raw_minimal":
        fill_numeric_median(tr, te, NUMERIC_COLS)
        fill_categorical_missing(tr, te, CATEGORICAL_COLS)
    elif variant == "processed_basic":
        add_missing_indicators(tr, te, NUMERIC_COLS + CATEGORICAL_COLS)
        fill_numeric_median(tr, te, NUMERIC_COLS)
        fill_categorical_missing(tr, te, CATEGORICAL_COLS)
```

```
    elif variant == "processed_extended":
        add_missing_indicators(tr, te, NUMERIC_COLS + CATEGORICAL_COLS)
        fill_numeric_median(tr, te, NUMERIC_COLS)
        fill_categorical_missing(tr, te, CATEGORICAL_COLS)
        add_extended_features(tr, te)

    feature_cols = [c for c in tr.columns if c not in {ID_COL, TARGET_COL}]
    cat_cols = finalize_frame_types(tr, te, feature_cols)
    meta = {"feature_cols": feature_cols, "cat_cols": cat_cols}
    return tr, te, meta


def get_feature_bundle(feature_variant: str):
    if feature_variant not in FEATURE_CACHE:
        FEATURE_CACHE[feature_variant] = build_features(
            feature_variant, train_df, test_df
        )
        print(f"[cache] built features for {feature_variant}")
    return FEATURE_CACHE[feature_variant]
```

[12]:
```
# Cell: Feature bundle preview - compare the three variants before training
feature_preview_rows = []
for variant in ["raw_minimal", "processed_basic", "processed_extended"]:
    tr_variant, te_variant, meta_variant = get_feature_bundle(variant)
    feature_preview_rows.append(
        {
            "variant": variant,
            "train_rows": len(tr_variant),
            "test_rows": len(te_variant),
            "feature_count": len(meta_variant["feature_cols"]),
            "categorical_feature_count": len(meta_variant["cat_cols"]),
        }
    )

feature_preview_df = pd.DataFrame(feature_preview_rows)
display(feature_preview_df)

preview_train, _, preview_meta = get_feature_bundle("processed_extended")
display(preview_train[preview_meta["feature_cols"][:12]].head())
```

```
[cache] built features for raw_minimal
[cache] built features for processed_basic
[cache] built features for processed_extended
```

```
              variant  train_rows  test_rows  feature_count  \
0         raw_minimal       32000       8000             11
1     processed_basic       32000       8000             22
2  processed_extended       32000       8000             41
```

```
   categorical_feature_count
0                          2
1                          2
2                          3

   no_of_dependents     education self_employed   income_annum  \
0            0.0000      Graduate            No 2,961,480.0000
1            2.0000  Not Graduate           Yes 2,832,292.0000
2            2.0000      Graduate            No 2,594,879.0000
3            4.0000      Graduate            No 2,497,120.0000
4            5.0000  Not Graduate           Yes 2,961,480.0000

      loan_amount  loan_term  cibil_score  residential_assets_value  \
0  7,271,300.0000     5.0000     622.0000            6,572,296.0000
1 11,746,169.0000     9.0000     482.0000            3,274,507.0000
2  8,265,934.0000    16.0000     786.0000              179,550.0000
3 26,116,812.0000    16.0000     352.0000           16,697,499.0000
4  4,071,241.0000    19.0000     381.0000            1,812,942.0000

   commercial_assets_value  luxury_assets_value  bank_asset_value  \
0          12,264,202.0000       9,802,424.0000    4,471,627.0000
1          16,079,677.0000       2,643,225.0000    3,021,958.0000
2           1,361,037.0000       9,057,816.0000    3,998,887.5000
3           4,531,977.0000               0.0000      983,214.0000
4           2,788,429.0000       7,977,004.0000   11,154,770.0000

   mis_no_of_dependents
0                0.0000
1                0.0000
2                0.0000
3                0.0000
4                0.0000
```

1.8 Fold Setup

We use stratified CV because the evaluation metric is F1 on a binary target, so keeping the class balance stable across folds is a good default.

[ ]:
```
# Cell: Fold setup - create stratified folds and save fold assignments
def make_folds(y: pd.Series, n_splits: int, seed: int):
    idx = np.arange(len(y))
    splitter = StratifiedKFold(n_splits=n_splits, shuffle=True,␣
 ↪random_state=seed)
    folds = []
    for tr_idx, va_idx in splitter.split(idx, y):
        folds.append((tr_idx, va_idx))
```

```
    fold_id = np.full(len(y), -1, dtype=np.int16)
    for fold, (_, va_idx) in enumerate(folds):
        fold_id[va_idx] = fold
    if (fold_id < 0).any():
        raise RuntimeError("Some rows were not assigned to folds.")
    return folds, fold_id


FOLDS, FOLD_ID = make_folds(train_df[TARGET_COL], N_SPLITS, SEED)
fold_index_path = ARTIFACT_DIR / f"fold_indices_{NOTEBOOK_SLUG}.csv"
pd.DataFrame({ID_COL: train_df[ID_COL], "fold_id": FOLD_ID}).to_csv(
    fold_index_path, index=False
)

display(pd.Series(FOLD_ID).value_counts().sort_index().rename("rows_per_fold"))
print("Saved fold assignments to", fold_index_path)
```

```
0    6400
1    6400
2    6400
3    6400
4    6400
Name: rows_per_fold, dtype: int64
```

```
Saved fold assignments to /kaggle/working/outputs/fold_indices_loan23-03-human-
eda-gpu-fast.csv
```

1.9 Modeling

This is the stronger part of the notebook. We keep the same CatBoost training flow from `loan23-02`, but show it more clearly:

- build the model
- fit each fold
- collect OOF and test predictions
- compare ablations
- search for the best threshold for F1

[ ]:
```
# Cell: Modeling helpers - train CatBoost, collect fold predictions, and␣
↪summarize experiments
def build_model(
    seed: int, params_override: dict | None = None, force_cpu: bool = False
):
    params_override = params_override or {}
    use_gpu = USE_GPU and GPU_AVAILABLE and not force_cpu
    base = {
        "loss_function": "Logloss",
        "eval_metric": "F1",
        "learning_rate": 0.03,
```

```
        "depth": 8,
        "l2_leaf_reg": 5.0,
        "iterations": CATBOOST_ITERATIONS,
        "random_seed": seed,
        "allow_writing_files": False,
        "verbose": False,
    }
    if use_gpu:
        base.update({"task_type": "GPU", "devices": GPU_DEVICE})
    base.update(params_override)
    return CatBoostClassifier(**base)


def fit_predict_one_fold(model, X_tr, y_tr, X_va, y_va, X_te, cat_cols:␣
 ↪list[str]):
    cat_idx = [X_tr.columns.get_loc(c) for c in cat_cols if c in X_tr.columns]
    model.fit(
        X_tr,
        y_tr,
        eval_set=(X_va, y_va),
        cat_features=cat_idx if cat_idx else None,
        use_best_model=True,
        early_stopping_rounds=EARLY_STOPPING_ROUNDS,
        verbose=False,
    )
    pred_va = model.predict_proba(X_va)[:, 1]
    pred_te = model.predict_proba(X_te)[:, 1]
    return pred_va, pred_te


def extract_feature_importance(model, feature_names: list[str]):
    try:
        vals = model.get_feature_importance()
        return pd.DataFrame({"feature": feature_names, "importance": vals})
    except Exception:
        return pd.DataFrame(columns=["feature", "importance"])


def threshold_search_rows(y_true: np.ndarray, pred_proba: np.ndarray,␣
 ↪experiment: str):
    rows = []
    thresholds = np.round(np.linspace(0.05, 0.95, 181), 4)
    for thr in thresholds:
        pred_label = (pred_proba >= float(thr)).astype(int)
        rows.append(
            {
                "experiment": experiment,
```

```
                "threshold": float(thr),
                "f1": float(f1_score(y_true, pred_label)),
                "precision": float(
                    precision_score(y_true, pred_label, zero_division=0)
                ),
                "recall": float(recall_score(y_true, pred_label,␣
 ↪zero_division=0)),
            }
        )
    return pd.DataFrame(rows)


def pick_best_threshold(search_df: pd.DataFrame):
    ranked = search_df.sort_values(
        ["f1", "precision", "recall", "threshold"],
        ascending=[False, False, False, True],
    ).reset_index(drop=True)
    return ranked.iloc[0]
```

[ ]:
```
# Cell: Run experiments - compare feature variants and record OOF/test␣
 ↪predictions
EXP_ROWS = []
OOF_STORE = {}
TEST_STORE = {}
FI_ROWS = []
THRESHOLD_ROWS = []


def run_experiment(exp_config: dict):
    name = exp_config["name"]
    feature_variant = exp_config["feature_variant"]
    params_override = exp_config.get("params_override") or {}
    seed = int(exp_config.get("seed", SEED))

    t0 = time.time()
    tr, te, meta = get_feature_bundle(feature_variant)
    feature_cols = meta["feature_cols"]
    cat_cols = meta["cat_cols"]

    X = tr[feature_cols].copy()
    X_test = te[feature_cols].copy()
    y = tr[TARGET_COL].to_numpy(dtype=int)

    oof_pred = np.zeros(len(tr), dtype=float)
    test_fold_preds = []
    fold_f1s = []
    fold_precisions = []
```

```
    fold_recalls = []
    force_cpu = False

    print(
        f"\n[{name}] start | variant={feature_variant} folds={len(FOLDS)}␣
↪device={TRAIN_DEVICE_LABEL}"
    )
    for fold, (idx_tr, idx_va) in enumerate(FOLDS):
        fold_t0 = time.time()
        X_tr = X.iloc[idx_tr].copy()
        y_tr = y[idx_tr]
        X_va = X.iloc[idx_va].copy()
        y_va = y[idx_va]

        model = build_model(
            seed + fold, params_override=params_override, force_cpu=force_cpu
        )
        try:
            pred_va, pred_te = fit_predict_one_fold(
                model, X_tr, y_tr, X_va, y_va, X_test, cat_cols
            )
        except Exception as exc:
            if not force_cpu and GPU_AVAILABLE:
                print(
                    f"[{name}] fold {fold + 1}/{len(FOLDS)} GPU failed␣
↪({type(exc).__name__}), retrying on CPU."
                )
                force_cpu = True
                model = build_model(
                    seed + fold, params_override=params_override, force_cpu=True
                )
                pred_va, pred_te = fit_predict_one_fold(
                    model, X_tr, y_tr, X_va, y_va, X_test, cat_cols
                )
            else:
                raise

        oof_pred[idx_va] = pred_va
        test_fold_preds.append(pred_te)

        pred_label_05 = (pred_va >= 0.5).astype(int)
        fold_f1 = float(f1_score(y_va, pred_label_05))
        fold_precision = float(precision_score(y_va, pred_label_05,␣
↪zero_division=0))
        fold_recall = float(recall_score(y_va, pred_label_05, zero_division=0))
        fold_f1s.append(fold_f1)
        fold_precisions.append(fold_precision)
```

```
        fold_recalls.append(fold_recall)

        fi = extract_feature_importance(model, feature_cols)
        if len(fi):
            fi["experiment"] = name
            fi["fold"] = fold
            FI_ROWS.append(fi)

        fold_elapsed = float(time.time() - fold_t0)
        print(
            f"[{name}] fold {fold + 1}/{len(FOLDS)} "
            f"f1@0.5={fold_f1:.6f} precision={fold_precision:.6f} "
            f"recall={fold_recall:.6f} time={fold_elapsed:.1f}s"
        )

    test_pred = np.mean(np.vstack(test_fold_preds), axis=0)
    search_df = threshold_search_rows(y, oof_pred, name)
    THRESHOLD_ROWS.extend(search_df.to_dict(orient="records"))
    best_thr_row = pick_best_threshold(search_df)

    oof_label_05 = (oof_pred >= 0.5).astype(int)
    oof_label_best = (oof_pred >= float(best_thr_row["threshold"])).astype(int)

    elapsed = float(time.time() - t0)
    device_used = "cpu-fallback" if force_cpu else TRAIN_DEVICE_LABEL
    EXP_ROWS.append(
        {
            "experiment": name,
            "model_family": "catboost",
            "feature_variant": feature_variant,
            "device": device_used,
            "best_threshold": float(best_thr_row["threshold"]),
            "cv_f1_mean_05": float(np.mean(fold_f1s)),
            "cv_f1_std_05": float(np.std(fold_f1s)),
            "cv_precision_mean_05": float(np.mean(fold_precisions)),
            "cv_recall_mean_05": float(np.mean(fold_recalls)),
            "oof_f1_05": float(f1_score(y, oof_label_05)),
            "oof_f1_best": float(f1_score(y, oof_label_best)),
            "oof_precision_best": float(
                precision_score(y, oof_label_best, zero_division=0)
            ),
            "oof_recall_best": float(recall_score(y, oof_label_best,␣
↪zero_division=0)),
            "elapsed_sec": elapsed,
        }
    )
    OOF_STORE[name] = oof_pred
```

```
    TEST_STORE[name] = test_pred
    print(
        f"[{name}] done | best_threshold={float(best_thr_row['threshold']):.4f}␣
↪"
        f"oof_f1_best={float(f1_score(y, oof_label_best)):.6f} time={elapsed:.
↪1f}s device={device_used}"
    )


BASE_EXPERIMENTS = [
    {"name": "catboost_processed_basic", "feature_variant": "processed_basic"},
    {"name": "catboost_processed_extended", "feature_variant":␣
↪"processed_extended"},
]

if ENABLE_ABLATIONS:
    BASE_EXPERIMENTS.insert(
        0, {"name": "catboost_raw_minimal", "feature_variant": "raw_minimal"}
    )

for exp in BASE_EXPERIMENTS:
    run_experiment(exp)

leaderboard_df = (
    pd.DataFrame(EXP_ROWS)
    .sort_values(["oof_f1_best", "cv_f1_mean_05"], ascending=[False, False])
    .reset_index(drop=True)
)
display(leaderboard_df)
```

```
[catboost_raw_minimal] start | variant=raw_minimal folds=5 device=gpu
[catboost_raw_minimal] fold 1/5 f1@0.5=0.704806 precision=0.545712
recall=0.994836 time=43.0s
[catboost_raw_minimal] fold 2/5 f1@0.5=0.704858 precision=0.544573
recall=0.998853 time=12.8s
[catboost_raw_minimal] fold 3/5 f1@0.5=0.705263 precision=0.544886
recall=0.999426 time=14.4s
[catboost_raw_minimal] fold 4/5 f1@0.5=0.704918 precision=0.544559
recall=0.999139 time=15.2s
[catboost_raw_minimal] fold 5/5 f1@0.5=0.705048 precision=0.545141
recall=0.997705 time=14.7s
[catboost_raw_minimal] done | best_threshold=0.0500 oof_f1_best=0.705240
time=101.6s device=gpu

[catboost_processed_basic] start | variant=processed_basic folds=5 device=gpu
[catboost_processed_basic] fold 1/5 f1@0.5=0.705429 precision=0.545255
```

```
recall=0.998853 time=13.6s
[catboost_processed_basic] fold 2/5 f1@0.5=0.705203 precision=0.544900
recall=0.999139 time=14.6s
[catboost_processed_basic] fold 3/5 f1@0.5=0.705394 precision=0.544957
recall=0.999713 time=18.4s
[catboost_processed_basic] fold 4/5 f1@0.5=0.704654 precision=0.544927
recall=0.996845 time=12.5s
[catboost_processed_basic] fold 5/5 f1@0.5=0.705454 precision=0.544943
recall=1.000000 time=12.8s
[catboost_processed_basic] done | best_threshold=0.4900 oof_f1_best=0.705254
time=73.6s device=gpu

[catboost_processed_extended] start | variant=processed_extended folds=5
device=gpu
[catboost_processed_extended] fold 1/5 f1@0.5=0.705406 precision=0.545056
recall=0.999426 time=14.1s
[catboost_processed_extended] fold 2/5 f1@0.5=0.704702 precision=0.544814
recall=0.997418 time=15.3s
[catboost_processed_extended] fold 3/5 f1@0.5=0.705454 precision=0.544943
recall=1.000000 time=15.7s
[catboost_processed_extended] fold 4/5 f1@0.5=0.705251 precision=0.544787
recall=0.999713 time=16.7s
[catboost_processed_extended] fold 5/5 f1@0.5=0.704870 precision=0.544672
recall=0.998566 time=13.9s
[catboost_processed_extended] done | best_threshold=0.0500 oof_f1_best=0.705240
time=77.3s device=gpu

                    experiment model_family     feature_variant device  \
0     catboost_processed_basic     catboost     processed_basic    gpu
1  catboost_processed_extended     catboost  processed_extended    gpu
2         catboost_raw_minimal     catboost         raw_minimal    gpu

   best_threshold  cv_f1_mean_05  cv_f1_std_05  cv_precision_mean_05  \
0          0.4900         0.7052        0.0003                0.5450
1          0.0500         0.7051        0.0003                0.5449
2          0.0500         0.7050        0.0002                0.5450

   cv_recall_mean_05  oof_f1_05  oof_f1_best  oof_precision_best  \
0             0.9989     0.7052       0.7053              0.5447
1             0.9990     0.7051       0.7052              0.5447
2             0.9980     0.7050       0.7052              0.5447

   oof_recall_best  elapsed_sec
0           1.0000      73.6077
1           1.0000      77.2578
2           1.0000     101.5884
```

1.10 Model Selection And Outputs

Now we lock in the best experiment, export the important artifacts, and build the submission file with the tuned threshold.

[16]:
```
# Cell: Save outputs - artifacts, OOF review tables, feature importance, and␣
↪submission CSV
results_df = (
    pd.DataFrame(EXP_ROWS)
    .sort_values(["oof_f1_best", "cv_f1_mean_05"], ascending=[False, False])
    .reset_index(drop=True)
)
display(results_df)

exp_path = ARTIFACT_DIR / f"experiments_{NOTEBOOK_SLUG}.csv"
results_df.to_csv(exp_path, index=False)

threshold_df = (
    pd.DataFrame(THRESHOLD_ROWS)
    .sort_values(["experiment", "f1", "threshold"], ascending=[True, False,␣
↪True])
    .reset_index(drop=True)
)
threshold_path = ARTIFACT_DIR / f"thresholds_{NOTEBOOK_SLUG}.csv"
threshold_df.to_csv(threshold_path, index=False)

best_exp = results_df.iloc[0]["experiment"]
best_row = results_df.iloc[0]
best_oof = OOF_STORE[best_exp]
best_test = TEST_STORE[best_exp]
best_threshold = float(best_row["best_threshold"])

oof_df = pd.DataFrame(
    {
        ID_COL: train_df[ID_COL],
        "y_true": train_df[TARGET_COL].astype(int),
        "pred_proba": best_oof,
        "pred_label_05": (best_oof >= 0.5).astype(int),
        "pred_label_best": (best_oof >= best_threshold).astype(int),
        "experiment": best_exp,
    }
)
for c in CATEGORICAL_COLS + NUMERIC_COLS:
    if c in train_df.columns and c not in oof_df.columns:
        oof_df[c] = train_df[c]

oof_path = ARTIFACT_DIR / f"oof_predictions_{NOTEBOOK_SLUG}.csv"
oof_df.to_csv(oof_path, index=False)
```

```python
if FI_ROWS:
    fi_df = pd.concat(FI_ROWS, ignore_index=True)
    fi_summary = (
        fi_df.groupby(["experiment", "feature"], as_index=False)["importance"]
        .mean()
        .sort_values(["experiment", "importance"], ascending=[True, False])
    )
else:
    fi_summary = pd.DataFrame(columns=["experiment", "feature", "importance"])

fi_path = ARTIFACT_DIR / f"feature_importance_{NOTEBOOK_SLUG}.csv"
fi_summary.to_csv(fi_path, index=False)

submission_df = pd.DataFrame(
    {
        ID_COL: test_df[ID_COL].astype(int),
        TARGET_COL: (best_test >= best_threshold).astype(int),
    }
)[[ID_COL, TARGET_COL]]
sub_path = OUTPUT_ROOT / f"submission_{NOTEBOOK_SLUG}.csv"
submission_df.to_csv(sub_path, index=False)

print("Saved experiments       :", exp_path)
print("Saved thresholds        :", threshold_path)
print("Saved OOF predictions   :", oof_path)
print("Saved feature importance:", fi_path)
print("Saved submission        :", sub_path)
print()
print("Recommended model :", best_exp)
print("OOF F1 @ 0.5      :", round(float(best_row["oof_f1_05"]), 6))
print("OOF F1 @ best     :", round(float(best_row["oof_f1_best"]), 6))
print("Best threshold    :", round(best_threshold, 4))
print("CV F1 mean @ 0.5  :", round(float(best_row["cv_f1_mean_05"]), 6))
print("CV F1 std @ 0.5   :", round(float(best_row["cv_f1_std_05"]), 6))
display(submission_df.head())
```

```
                    experiment model_family     feature_variant device  \
0     catboost_processed_basic     catboost     processed_basic    gpu
1  catboost_processed_extended     catboost  processed_extended    gpu
2         catboost_raw_minimal     catboost         raw_minimal    gpu

   best_threshold  cv_f1_mean_05  cv_f1_std_05  cv_precision_mean_05  \
0          0.4900         0.7052        0.0003                0.5450
1          0.0500         0.7051        0.0003                0.5449
2          0.0500         0.7050        0.0002                0.5450

   cv_recall_mean_05  oof_f1_05  oof_f1_best  oof_precision_best  \
```

```
0             0.9989     0.7052       0.7053              0.5447
1             0.9990     0.7051       0.7052              0.5447
2             0.9980     0.7050       0.7052              0.5447

   oof_recall_best  elapsed_sec
0           1.0000      73.6077
1           1.0000      77.2578
2           1.0000     101.5884
Saved experiments       : /kaggle/working/outputs/experiments_loan23-03-human-
eda-gpu-fast.csv
Saved thresholds        : /kaggle/working/outputs/thresholds_loan23-03-human-
eda-gpu-fast.csv
Saved OOF predictions   :
/kaggle/working/outputs/oof_predictions_loan23-03-human-eda-gpu-fast.csv
Saved feature importance:
/kaggle/working/outputs/feature_importance_loan23-03-human-eda-gpu-fast.csv
Saved submission        : /kaggle/working/submission_loan23-03-human-eda-gpu-
fast.csv

Recommended model : catboost_processed_basic
OOF F1 @ 0.5      : 0.705227
OOF F1 @ best     : 0.705254
Best threshold    : 0.49
CV F1 mean @ 0.5  : 0.705227
CV F1 std @ 0.5   : 0.0003

      id  loan_status
0  32001            1
1  32002            1
2  32003            1
3  32004            1
4  32005            1
```

[ ]:
```
# Cell: Error review - inspect confusion matrix, high-confidence misses, and␣
↪feature importance
best_pred_label = (best_oof >= best_threshold).astype(int)
cm = confusion_matrix(train_df[TARGET_COL].astype(int), best_pred_label)
cm_df = pd.DataFrame(cm, index=["actual_0", "actual_1"], columns=["pred_0",␣
↪"pred_1"])
display(cm_df)

error_review_df = oof_df.copy()
error_review_df["error"] = (
    error_review_df["y_true"] != error_review_df["pred_label_best"]
).astype(int)
error_review_df["confidence"] = np.where(
    error_review_df["pred_label_best"] == 1,
```

```
    error_review_df["pred_proba"],
    1.0 - error_review_df["pred_proba"],
)

hard_errors = (
    error_review_df[error_review_df["error"] == 1]
    .sort_values("confidence", ascending=False)
    .head(10)
)
print("Top high-confidence mistakes:")
display(hard_errors)

if len(fi_summary):
    top_features = fi_summary[fi_summary["experiment"] == best_exp].head(20)
    display(top_features)
    plt.figure(figsize=(10, 8))
    sns.barplot(data=top_features, x="importance", y="feature",␣
 ↪palette="viridis")
    plt.title(f"Top feature importance - {best_exp}")
    plt.tight_layout()
    plt.show()
```

```
          pred_0  pred_1
actual_0       1   14569
actual_1       0   17430

Top high-confidence mistakes:

          id  y_true  pred_proba  pred_label_05  pred_label_best  \
19636  19637       0      0.6049              1                1
8905    8906       0      0.5817              1                1
11590  11591       0      0.5815              1                1
544      545       0      0.5797              1                1
16653  16654       0      0.5776              1                1
24433  24434       0      0.5776              1                1
1889    1890       0      0.5768              1                1
17697  17698       0      0.5764              1                1
20230  20231       0      0.5744              1                1
28419  28420       0      0.5737              1                1

                     experiment      education self_employed  \
19636  catboost_processed_basic       Graduate           NaN
8905   catboost_processed_basic       Graduate           NaN
11590  catboost_processed_basic       Graduate           NaN
544    catboost_processed_basic            NaN           NaN
16653  catboost_processed_basic       Graduate            No
24433  catboost_processed_basic       Graduate           Yes
1889   catboost_processed_basic   Not Graduate           NaN
17697  catboost_processed_basic   Not Graduate           Yes
```

20230	catboost_processed_basic	Not Graduate	NaN
28419	catboost_processed_basic	Graduate	NaN

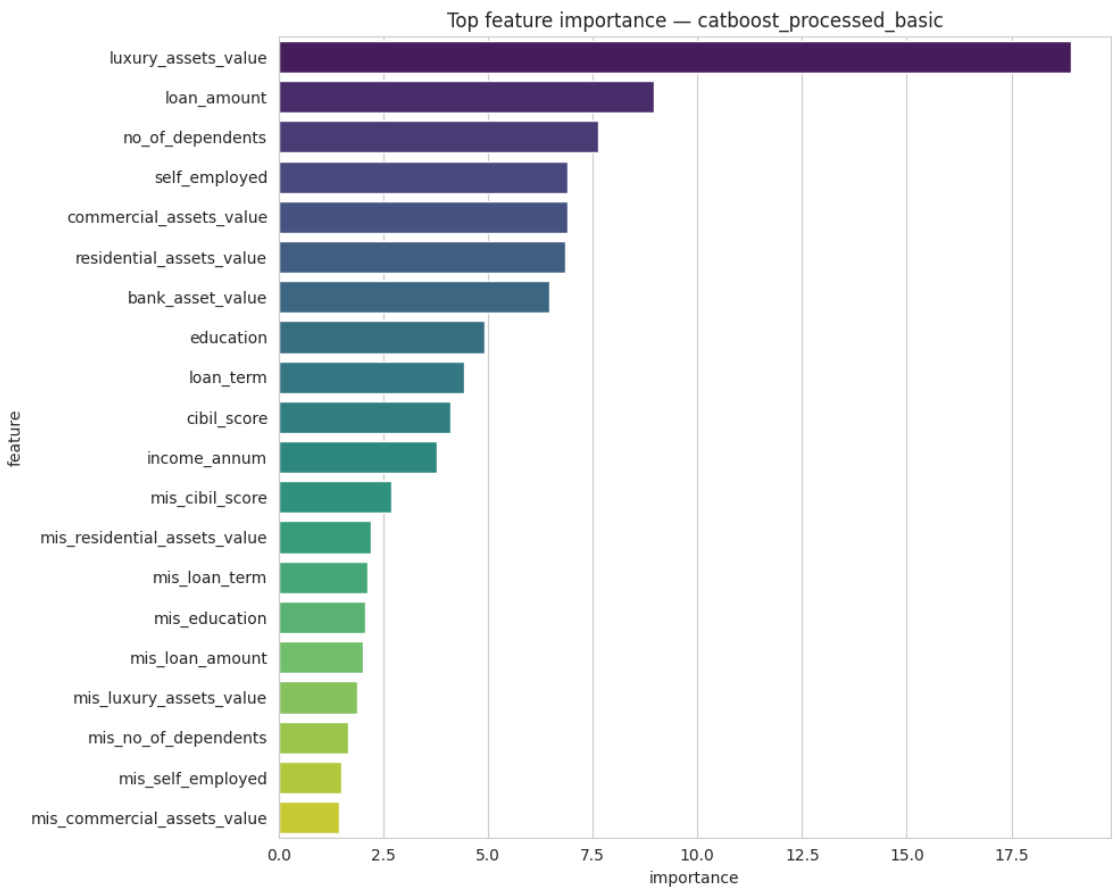
	no_of_dependents	income_annum	loan_amount	loan_term	\
19636	2.0000	7,226,188.0000	0.0000	20.0000	
8905	4.0000	6,348,007.0000	12,506,267.0000	4.0000	
11590	4.0000	5,529,012.0000	2,861,576.0000	2.0000	
544	5.0000	1,476,805.0000	178,272.0000	12.0000	
16653	3.0000	7,409,572.0000	0.0000	6.0000	
24433	0.0000	181,652.0000	0.0000	21.0000	
1889	2.0000	1,234,984.0000	0.0000	18.0000	
17697	2.0000	0.0000	1,823,435.0000	18.0000	
20230	2.0000	7,513,986.0000	3,317,418.0000	20.0000	
28419	4.0000	312,598.0000	1,590,405.0000	3.0000	

	cibil_score	residential_assets_value	commercial_assets_value	\
19636	422.0000	0.0000	18,288,092.0000	
8905	NaN	592,740.0000	23,435,164.0000	
11590	630.0000	0.0000	4,425,757.0000	
544	872.0000	618,059.0000	13,490,511.0000	
16653	NaN	0.0000	6,237,318.0000	
24433	519.0000	0.0000	18,845,739.0000	
1889	422.0000	0.0000	8,084,947.0000	
17697	798.0000	0.0000	NaN	
20230	883.0000	763,011.0000	7,408,802.0000	
28419	563.0000	14,484,219.0000	NaN	

	luxury_assets_value	bank_asset_value	error	confidence
19636	NaN	5,014,165.0000	1	0.6049
8905	96,800.0000	14,483,547.0000	1	0.5817
11590	0.0000	6,185,172.0000	1	0.5815
544	4,946,080.0000	3,325,209.0000	1	0.5797
16653	0.0000	6,509,972.0000	1	0.5776
24433	12,152,220.0000	5,807,010.0000	1	0.5776
1889	8,397,550.0000	14,107,683.0000	1	0.5768
17697	0.0000	5,742,412.0000	1	0.5764
20230	0.0000	1,249,865.0000	1	0.5744
28419	NaN	0.0000	1	0.5737

	experiment	feature	importance
7	catboost_processed_basic	luxury_assets_value	18.9282
5	catboost_processed_basic	loan_amount	8.9493
19	catboost_processed_basic	no_of_dependents	7.6326
21	catboost_processed_basic	self_employed	6.9111
2	catboost_processed_basic	commercial_assets_value	6.8991
20	catboost_processed_basic	residential_assets_value	6.8535
0	catboost_processed_basic	bank_asset_value	6.4613
3	catboost_processed_basic	education	4.9117

```
6   catboost_processed_basic                    loan_term      4.4210
1   catboost_processed_basic                  cibil_score      4.0920
4   catboost_processed_basic                 income_annum      3.7751
9   catboost_processed_basic              mis_cibil_score      2.6851
17  catboost_processed_basic  mis_residential_assets_value      2.2130
14  catboost_processed_basic                mis_loan_term      2.1087
11  catboost_processed_basic                mis_education      2.0753
13  catboost_processed_basic              mis_loan_amount      2.0154
15  catboost_processed_basic       mis_luxury_assets_value      1.8876
16  catboost_processed_basic          mis_no_of_dependents      1.6589
18  catboost_processed_basic             mis_self_employed      1.4823
10  catboost_processed_basic   mis_commercial_assets_value      1.4296
```



1.11 Submission

The last cell keeps Kaggle-native submission behavior. If credentials are available and `RUN_SUBMISSION` is enabled, the notebook will submit the CSV we just created.

[18]:
```
# Cell: Submission - send the final CSV to Kaggle when runtime flags allow it
if RUN_SUBMISSION:
    if api is None:
        raise RuntimeError("RUN_SUBMISSION=True but Kaggle API auth is not␣
 ↪available")

    submit_message = (
        f"{NOTEBOOK_SLUG} | "
        f"cv_mode={CV_MODE} n_splits={N_SPLITS} seed={SEED} | "
        f"time={datetime.now(UTC).isoformat()}"
    )
    api.competition_submit(
        file_name=str(OUTPUT_ROOT / f"submission_{NOTEBOOK_SLUG}.csv"),
        message=submit_message,
        competition=COMPETITION,
    )
    print("Submission sent:", submit_message)
else:
    print("RUN_SUBMISSION is False, so the submission file was created but not␣
 ↪sent.")
```

```
100%|      | 62.5k/62.5k [00:00<00:00, 159kB/s]

Submission sent: loan23-03-human-eda-gpu-fast | cv_mode=stratified n_splits=5
seed=42 | time=2026-03-21T14:21:23.419426+00:00
```